

Python Pattern and String Processing Assignment

Student Details

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Program 1: Pyramid Pattern Generator

Problem Statement

Create a Python function to generate and print a pyramid pattern of asterisks (*) with a user-provided height, ensuring the base of the pyramid has a width equal to the height.

Code

```
def pyramid(h):  
    for i in range(1, h + 1):  
        spaces = " " * (h - i)  
        stars = "*" * i  
        print(spaces + stars)
```

pyramid(5)

Concepts Used

- Functions (def)
- Parameters and Arguments
- For Loop
- String Multiplication
- Pattern Printing
- Indentation

What I Learned

- How to create and call functions.
- How to use loops to generate patterns.
- How to control spaces and symbols dynamically.
- How string multiplication works in Python.

Sample Output

```
*  
  
**  
  
***  
  
****  
  
*****
```

Program 2: Pattern Occurrence Counter

Problem Statement

Develop a Python program to recognize and count the occurrences of a specific sequence pattern (e.g., "abc") within a given string.

Code

```
def count_pattern(text, pattern):  
    count = 0  
    for i in range(len(text) - len(pattern) + 1):  
        if text[i:i+len(pattern)] == pattern:  
            count += 1  
    return count  
  
string = input("Enter a string: ")  
pattern = input("Enter a pattern: ")  
  
print("Occurrences:", count_pattern(string, pattern))
```

Concepts Used

- Functions
- User Input
- String Slicing
- Conditional Statements (if)
- For Loop
- Length Function (len())
- Counter Variable

What I Learned

- How to take input from users.
- How to compare strings.
- How slicing works in Python.
- How to count matching patterns in a string.
- How to use a counter variable effectively.

Sample Input

Enter a string: abcxyzabcabc

Enter a pattern: abc

Sample Output

Occurrences: 3

Program 3: Checkerboard Pattern Generator

Problem Statement

Write a Python script to generate and display a checkerboard pattern of a specified size, using alternating 'X' and 'O' characters for the squares.

Code

```
def checkerboard(size):  
    for i in range(size):  
        for j in range(size):  
            if (i + j) % 2 == 0:  
                print("X", end=" ")  
            else:  
                print("O", end=" ")  
        print()
```

```
checkerboard(5)
```

Concepts Used

- Functions
- Nested Loops
- Conditional Statements

- Modulus Operator (%)
- Pattern Printing
- Matrix Traversal Logic

What I Learned

- How nested loops work.
- How rows and columns are generated.
- How the modulus operator helps in alternating patterns.
- How to create matrix-style outputs.
- How to combine conditions with loops.

Sample Output

```
X O X O X
O X O X O
X O X O X
O X O X O
X O X O X
```

Overall Learning Outcomes

Through this assignment, I learned:

1. Function creation and function calling.
2. User input handling.
3. String manipulation and slicing.
4. Pattern generation using loops.
5. Nested loop implementation.
6. Conditional statements.
7. Counter logic.
8. Modulus operator usage.
9. Problem-solving using Python.
10. Writing structured and reusable code.

Conclusion

This assignment helped me understand Python fundamentals such as functions, loops, conditions, string operations, pattern generation, and problem-solving techniques. These concepts form the foundation for developing more advanced Python applications.